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Modeling Human Activity States Using Hidden Markov Models

1. Background and motivation

Human activity recognition using smartphone sensors is a powerful tool that can lead to many valuable applications [1][2]. In this project, I developed a Hidden Markov Model (HMM) using sensor activities from recorded accelerometer (x, y, z) and gyroscope (x, y, z) signals across four states (Still, Standing, Jumping, Walking). The HMM can be deployed to provide low-cost safety tracking for a wildlife ranger patrol monitoring system across four states: still (stationary ambush or emergency collapse), standing (lookout posture), walking (active patrol gait), and jumping (obstacle clearance or threat evasion) in African wildlife regions, enhancing worker safety in resource-constrained African environments.

Table 1*: Submission Links*

Deliverable	Link
GitHub Repository	https://github.com/hd77alu/human_sensor_activity_hmm
Data Collection Video	https://youtu.be/RwBWDiKYjbc?si=7daKXft7bgAq96iP

2. Data collection and pre-processing

2.1 Data Collection

I collected 50 samples of training data and 8 samples of test data with a 100 Hz sampling rate using the Sensory Logger mobile app. The training data contains approximately 12 samples per activity, collected across two separate sessions (morning and evening) to ensure that we have a sufficient amount of training data with realistic motion variance, which helps prevent overfitting. Testing data of 2 samples per activity was collected in a different environment (different room) with a different device than the one used to record the training

data; these testing samples were later used to measure model generalization during evaluation.

2.2 Data Preprocessing

Table 2*: Data Pre-Processing Step***s**

Preprocessing Step	Description
Session Data Loading	Reads continuous 3-axis accelerometer, 3-axis gyroscope, and metadata CSV files from separate directories, identifying activity labels based on folder names.
Attribute Renaming	Renames sensor-specific columns to prevent name collisions by adding prefixes such as (acc_)and (gyro_) to spatial coordinate columns.
Temporal Sorting	Sorts both accelerometer and gyroscope data streams by their epoch timestamps to ensure sequential integrity.
Timestamp Synchronization	Merges the independent high-frequency sensor streams using nearest-neighbor matching (pd.merge_asof) on timestamps to accommodate minor sampling offsets.

3. HMM setup and implementation details

I developed a continuous-emission Hidden Markov Model (HMM) from scratch with NumPy with four hidden states to capture the four physical activities as a sequence of hidden dynamical states emitting continuous sensor vectors. The Observation Space is composed of a continuous 5-dimensional feature vector extracted from the synchronized accelerometer and gyroscope time windows. The vector comprises the final selected features: four time-domain features and one frequency-domain feature, as outlined in Table 3. The emission model is characterized by Multivariate Gaussian distributions with diagonal covariance matrices to manage feature dispersion per state.

The model leverages the Log-Space Viterbi algorithm to find the most likely sequence of hidden activities, taking advantage of the algorithm's use of dynamic programming over the entire sequence to enforce physical state persistence, reflecting the reality that human activities (such as walking or standing) persist continuously over consecutive time windows [3][4].

The sensory sequence was segmented using a window-based feature extraction pipeline with a sliding window of 2 seconds (200 samples per window) and a 50% overlap to ensure smooth transitions and maintain temporal dependency between sequential states. For each window, thirty-eight time-domain features (e.g., mean, standard deviation, variance, root-mean-square (RMS), minimum, and maximum values) and seven Frequency-Domain Features (e.g., spectral energy and the dominant frequency component) were extracted. The top features were ranked using the ANOVA F-Score [5], then I selected the final features by retaining the accelerometer and gyroscope top four time-domain features and the top FFT frequency-domain feature to model the HMM, as outlined in the table below:

Table 3: *Feature Extraction & Selection Summary*

Feature Name	**Feature Domain**	**ANOVA F-Score**	**Justification for Selection**
Signal Magnitude Area (acc_sma)	Time Domain	1939.50	Measures total Signal Magnitude Area (linear kinetic energy scale). Establishes baseline physical intensity hierarchy.
Z-Axis Accelerometer Standard Deviation (acc_z_std)	Time Domain	1517.49	Measures linear acceleration dispersion along the *Z*Z-axis. Captures forward/backward gait impacts and vertical landing force fluctuations during step and jump cycles.

X-Axis Gyroscope Root Mean Square (gyro_x_rms)	Time Domain	1309.86	Measures Root Mean Square angular velocity along the X Z-axis. Quantifies overall rotational signal power to differentiate gross body movement from static posture.
X-Axis Gyroscope Standard Deviation (gyro_x_std)	Time Domain	1305.71	Measures angular velocity AC dispersion along the XZ-axis. Detects subtle involuntary postural sways and hand micro-tremors during Standing, distinguishing it from zero-rotation surface placement (Still).
Accelerometer Magnitude Spectral Energy (acc_mag_spectral_energy)	Frequency Domain	415.47	Quantifies total spectral power density derived from Fast Fourier Transform (FFT). Differentiates rhythmic gait periodicities (Walking at $\sim 1.5 - 2.0Hz$) from high-energy impact periodicities (Jumping at $\sim 2.5 - 3.5Hz$).

4. Results

The HMM model performance was evaluated on the unseen test set and yielded the following results:

Table 4: *Evaluation results*

State (Activity)	Number of Samples	Sensitivity (Recall)	Specificity	Overall Accuracy
Jumping	10	1.0000	1.0000	1.0000
Standing	11	1.0000	1.0000	1.0000
Still	9	1.0000	1.0000	1.0000
Walking	9	1.0000	1.0000	1.0000

Figure 1: *Transition Matrix Heatmap & Sequence Transition*

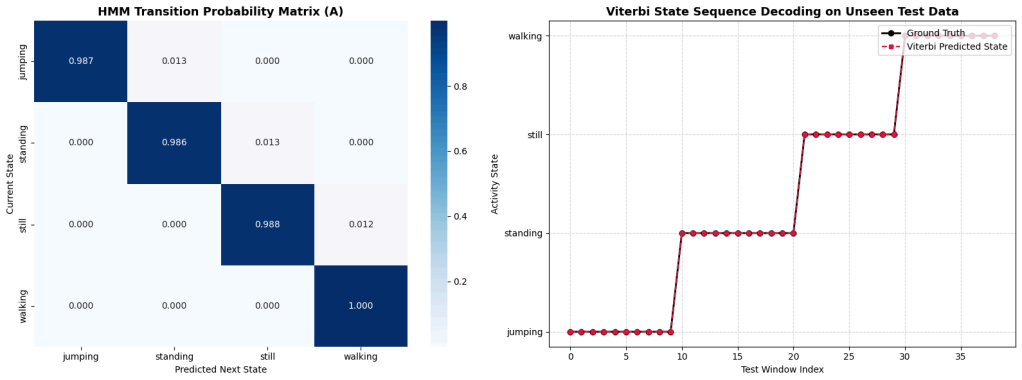
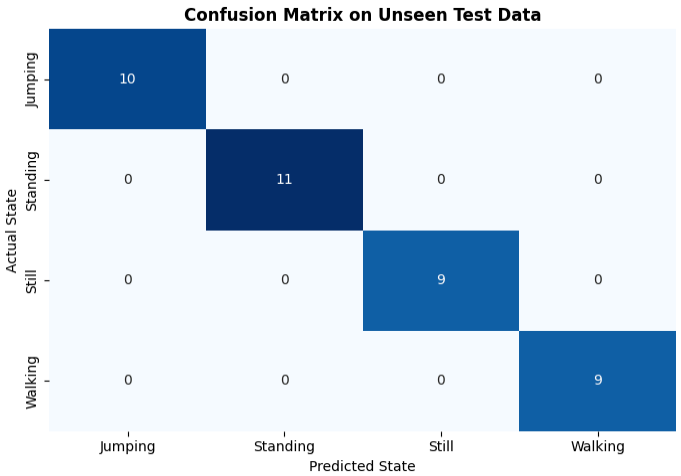


Figure 2: *Confusion Matrix*



5. Discussion and conclusion

The systematic feature selection approach (Table 3) that I followed has enabled the model to identify all of the activities accurately (Table 4) with no confusion (Figures 1- 2) between the four categories, demonstrating a strong ability to generalize across continuous sensor streams and achieving perfect sensitivity and specificity on unseen test sessions.

In conclusion, this project successfully developed and validated an end-to-end Human Activity Recognition framework utilizing a custom continuous-emission Gaussian Hidden Markov Model (HMM) built from scratch. These findings from the HMM architecture could be mapped directly to Wildlife Ranger Patrol Monitoring across national parks and game reserves in wildfire regions in Africa.

References

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